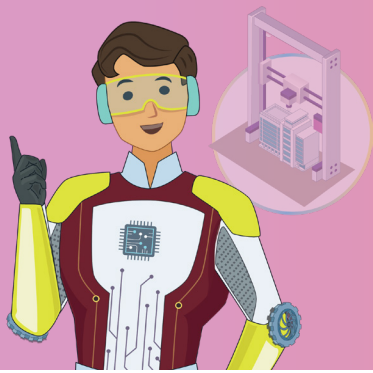




3D Printing

03

Every single object we interact with is either machine-made or handmade. Let me show you how different products are manufactured.



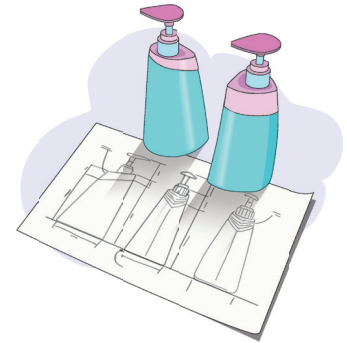
Every single product we use in our life is manufactured somewhere. It is either made by hand or by machine. Over the years, manufacturing techniques have changed and new techniques have been introduced to increase efficiency.

Before manufacturing, every product goes through the phases of ideation and prototyping.

Ideation: In this phase, the purpose, usability and the probable design of the product is discussed and finalised.

Prototyping: After ideation, a sample prototype of the product is created either on a software or with a sample model of the product.

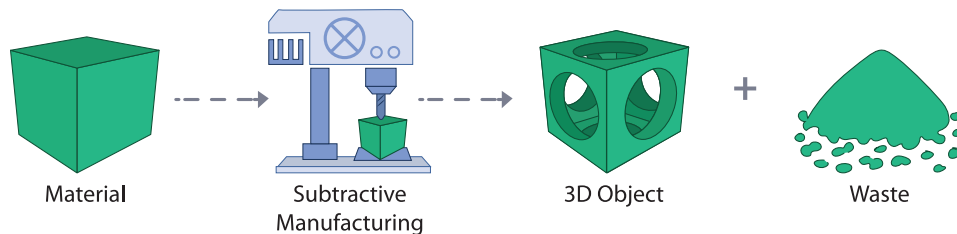
Manufacturing: Once the design of the product is finalised, it is manufactured using machines or by hand.



A product can be manufactured in multiple different ways. Among them, two methods are widely used:

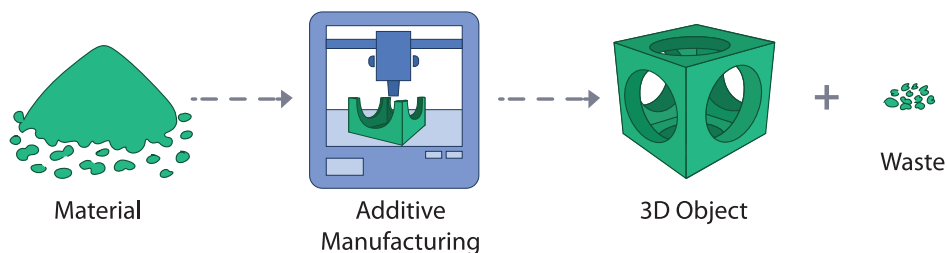
1. Subtractive Manufacturing (drilling, cutting, CNC milling and turning)

In this process, a piece of solid material (typically called stock) is placed in the work area. The machine removes unwanted material to create the final object. The main machine in this category is the mill, but others - including the lathe - are used.



2. Additive Manufacturing

In this process, the machine starts with an empty work area and typically extrudes or melts material onto it to create an object. The most visible machine in this category is a 3D printer.



Bytes

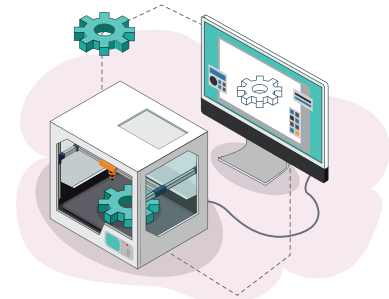
A lathe is a machine tool that shapes a piece of metal or wood by rotating it and removing unwanted material from it.

3. 3D Printing

Additive Manufacturing	Subtractive Manufacturing
In additive manufacturing, material is added layer by layer, one over the other to develop a solid 3D product.	In subtractive manufacturing, material is gradually removed, layer by layer, from a solid block to fabricate a 3D product.
This manufacturing concept is usually suitable for materials with a low melting point, such as plastic.	This manufacturing concept can be applied to all solid materials, irrespective of their melting point.
The weight of the constructive material of final component can be controlled during operation.	Material density cannot be controlled during operation. Density of an object remains the same as that of the initial solid block (usually a cast product).
No material wastage occurs in these processes.	These processes are associated with material wastage in the form of chips, scraps, dissolved ions, vapours, etc.
Complex shapes can be easily fabricated using additive manufacturing techniques.	Subtractive manufacturing processes have limited capability in fabrication of complex shapes.
Structures containing completely closed internal hollow parts can be produced by these processes.	Structures containing enclosed hollow parts cannot be produced by these processes, unless joining is allowed.
These processes are efficient and can provide superior quality and desired properties, without requiring any further processing.	These are usually suitable for mass production where the product quality requirement is not so rigid.

Each method has its own purposes. In this lesson, we will address the additive manufacturing technique.

3D printing is the process of creating a physical object using a 3D model.



Traditional printers require a cartridge to print on a surface, whereas a 3D printer prints a 3D object using a spool of special material.

Note:

A **spool** is a round object onto which thread, tape, or film can be wound, especially before it is put into a machine.

Earlier, 3D printing was predominately used for plastics, but now metal 3D printing is becoming a standard manufacturing process for industries like aerospace.

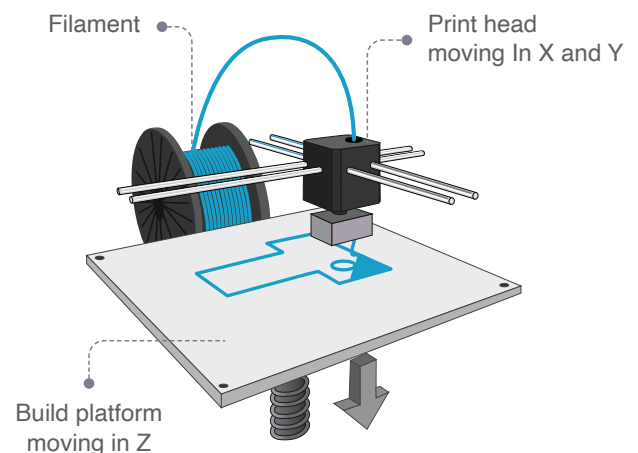
Plastic materials used are Nylon, PLA, ABS, ULTEM, ASA, TPU, and resins. The metals used in manufacturing are aluminium, stainless steel and titanium. Other 3D printing materials include composites, ceramics, wax, and sand.

Materials that are traditionally difficult to work with can be 3D printed.

Methods of 3D printing

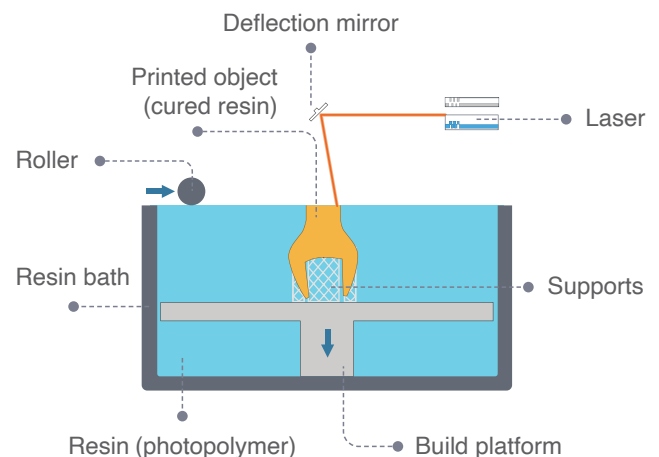
1. FDM (Fusion Deposition Modeling)

A spool of filament is first loaded into the printer. This material is thermoplastic, meaning it can be melted at a certain temperature and solidified when cooled. The nozzle of the printer is heated to a certain temperature and this filament is fed to the extrusion head and in the nozzle. The filament melts with the nozzle's temperature. The extrusion head is attached to a 3-axes system that allows it to move in X, Y and Z directions. The melted material is forced out in thin strands and is deposited layer-by-layer in predetermined locations, where it cools and solidifies.



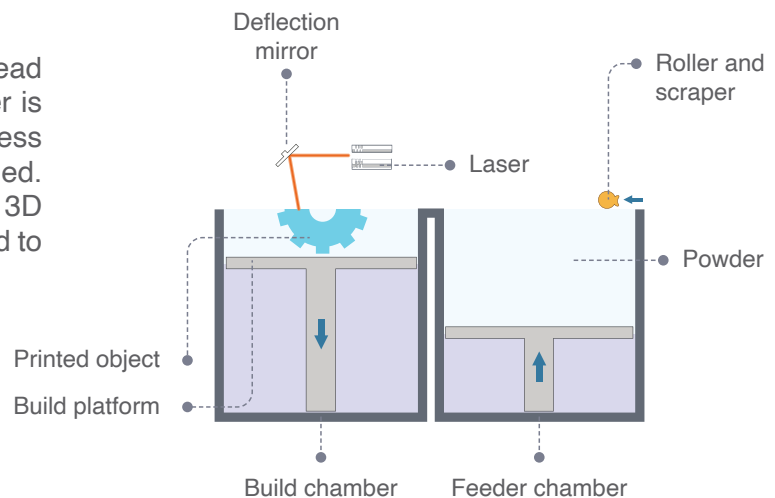
2. SLA (Stereolithography Apparatus)

SLA 3D printers use light-reactive materials called "resin." When SLA resins are exposed to certain wavelengths of light, they are converted into solid geometrical shapes which are either rigid or flexible. SLA parts have the highest resolution and accuracy, the sharpest details, and the smoothest surface finishes of all 3D printing technologies.



3. Selective Laser Sintering

In this technique, a super fine powder is spread across the platform. Then, a high-powered laser is flashed over the powder to solidify it. This process is repeated until the required 3D shape is formed. This process is not common in consumer level 3D printers, but on the industrial side, it is often used to create large durable objects out of metal.



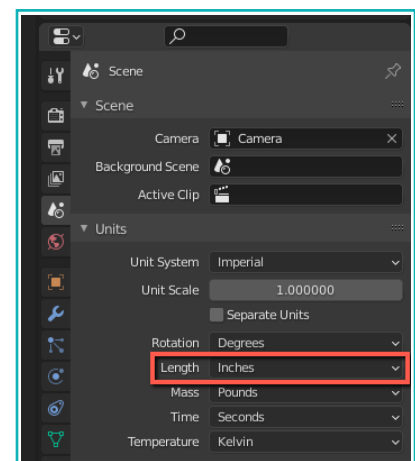
Apart from these, there are also more specialised 3D printing methods.

Now that we know how 3D printers work, let's learn how to prepare an object for 3D printing.

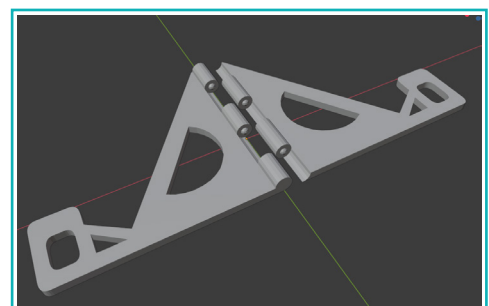
Recall the three main steps in any product formation; ideation, prototype and manufacturing. Ideation and prototyping are done either through physical Modelling or on software. One such software is Blender.

Let's see how we can use Blender to prototype a 3D object.

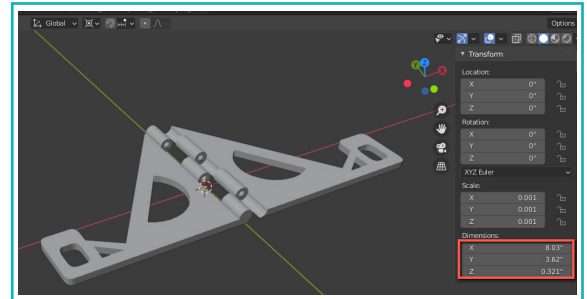
- 1 Make sure the units used to create the model are set correctly. Any 3D model we create, needs to have appropriate dimensions. Here, we have used inches.



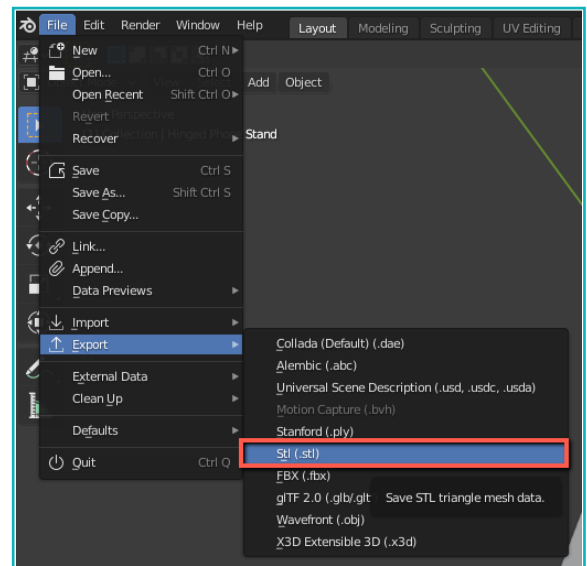
- 2 Make sure the designed 3D object lays flat. This is to ensure that the filament gets sufficient support during printing. It also makes sure there is only minimum wastage. We will create a pocket mobile stand with hinges.



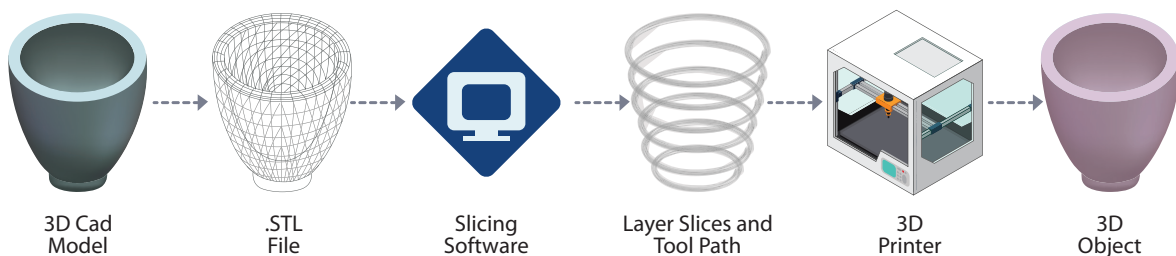
- 3 Double check to make sure the object is correctly sized.



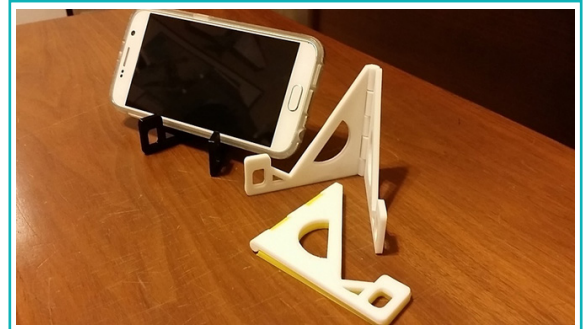
A 3D printing software supports several file formats. The most commonly used file format is .stl.



This file can be opened in any 3D printing software, depending on the 3D printer used. A 3D printing software reads the .stl file and slices it in layers to prepare it for printing.



- 4 Once the 3D printing process is complete, the object is assembled and is ready for use.



Exercise

 Tick mark ☒ the correct answer

- 1 Which of the following materials cannot be used for 3D printing?
- a. Plastics: nylon, PLA, ABS, ULTEM, ASA, TPU, resins ☐
 - b. Metals: aluminium, stainless steel, titanium ☐
 - c. Composites, ceramics, wax and sand ☐
 - d. Cotton, wool, silk ☐
- 2 Which of the following is not a 3D printer?
- | | | | |
|--------|--------------------------|------------------------------|--------------------------|
| a. FDM | ☐ | b. Canon color printer | ☐ |
| c. SLA | ☐ | d. Selective laser sintering | ☐ |

Applications of 3D printing

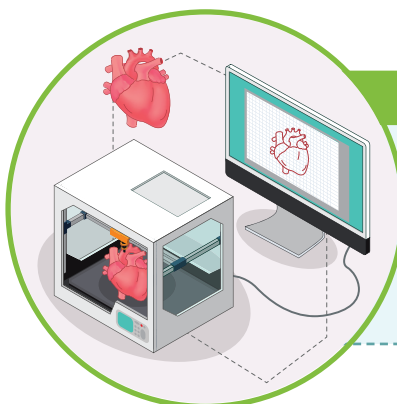
3D printing is used in sectors like architecture and construction, maritime industries, healthcare and medicine, chemical industries, mechanics, food industries, education, aeronautics and space, automotive. It is also applicable in textile and fashion, robotics, optics, electronics, and entertainment and broadcast industries.

3D printing can save our time and labour.
It can also help us build huge structures.



Tech Flash

- **3D printing** : 3D printing is the process of creating a physical object using a 3D model.
- **Subtractive Manufacturing** : In this process, a piece of solid material (typically called stock) is placed in the work area, and the machine removes material to create the object.
- **Additive Manufacturing** : In this process, the machine starts with an empty work area and typically extrudes or melts material onto it to create the object.



Future Tech

Bioprinting

In the future, 3D printers will also be used to create human tissue and organs. This will need medical and engineering experts to collaborate.

